2027 Guide to AI in Science

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2026-08-31

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Preface

AI is radically redefining the nature of scientific inquiry — for better and for worse.

In Fall 2026, faculty, graduate students, and postdocs across multiple departments at the University of Florida came together to understand this urgent development. In a seminar titled *Collective [Artificial] Intelligence in Science*, we documented and discussed how AI is changing each component of our research workflow. This includes project management, idea development, data analysis, writing, and dissemination. To push our understanding, we alternated between the perspective of an AI zealot and an AI critic – identifying key resources, consensus statements, and areas of disagreement along the way. The result is a *human-curated and written* **2027 Guide to AI in Science**.

The seminar began with an introduction to big-team science – its rise[1], competitive advantages[2], and relevance to the current moment. For instance, the paper describing OpenAI's GPT-1 had 4 authors; the paper describing GPT-3 had 31 authors; and the paper describing GPT-5 had 483 authors (Figure 1). Similarly, a prominent AI-led cybersecurity attack of the online community, Hugging Face, involved coordinated action between thousands of AI agents[3]. After reviewing these developments, the course itself became a large-scale collaboration, dedicated to understanding how AI is changing the nature of science.

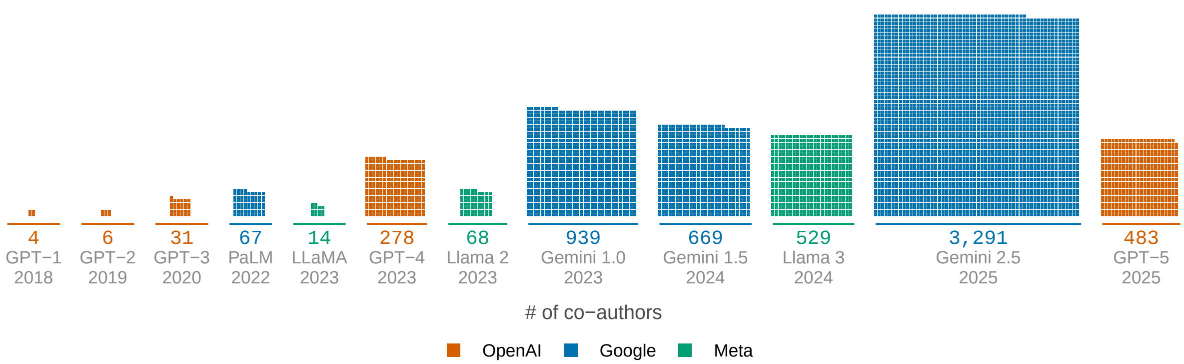

Figure 1: Credited co-authors on flagship AI model reports from OpenAI, Google, and Meta, 2018–2025. One square is one credited person; each block's area is its author list.

Each week followed a consistent structure (Figure 2). First, we would select a component of the research pipeline to examine (e.g., data analysis). Next, we would spend one week compiling resources for understanding the emerging role of AI in that part of the research pipeline. In class, we engaged in peer-to-peer teaching – all with the goal of identifying the top 3 resources to include in the guide. Last, we used a live onlined voting platform to solicit, debate, and vote upon key takeaways for the week (https://pol.is/). The result is the **2027 Guide to AI in Science**: a summary of our top resources, takeaways, and disagreements.

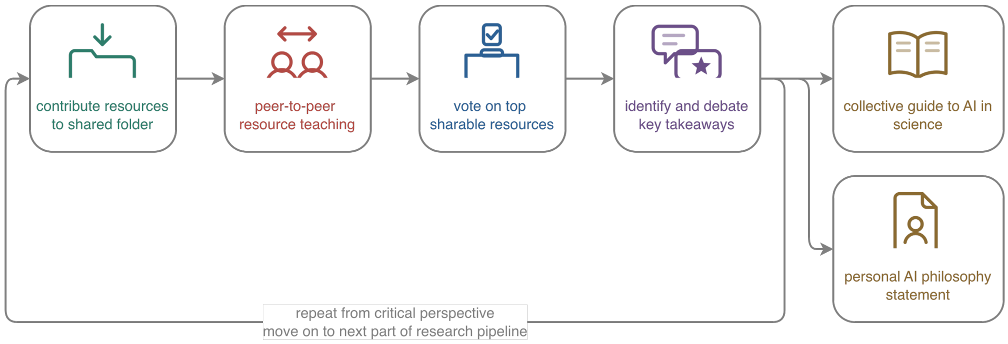

Figure 2: Structure of the University of Florida Fall 2026 seminar, titled Collective [Artificial] Intelligence in Science.

Ideation and Project Management

Ideation and Project Management: Using AI

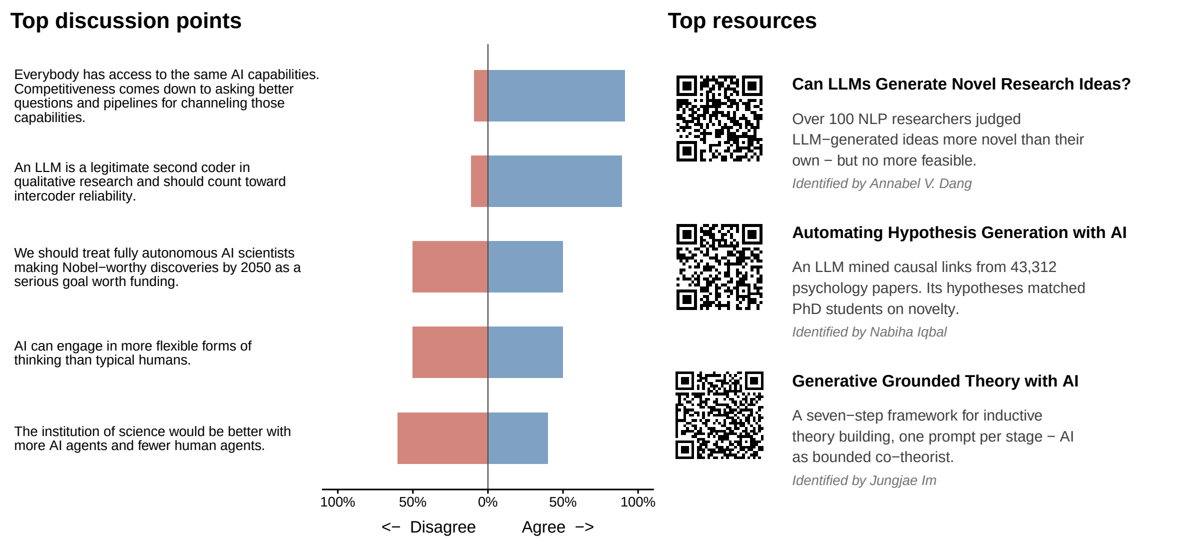

Figure 1: Overview of Week 2 of the Fall 2026 Collective [Artificial] Intelligence Seminar. Left: top discussion points, along with estimated level of agreement among seminar participants. Right: three most popular resources identified by seminar participants.

Many researchers' first contact with AI tools is in the realm of ideation or project management. For example, one seminar participant noted that they have their research trainees submit weekly project updates to a repository shared with remote deployment of the LLM, Claude Opus (Figure 2, panel A). This allowed them to later prompt the model with questions about past conversations, challenges, decisions, and solutions (Figure 2, panel B). This same seminar participant noted that this allowed them to efficiently take stock of the trainee's progress, have a conversation with the LLM about potential future directions, and generate a simulated figure to send back to the trainee to convey next steps (Figure 2, panel C).

As illustrated by the most popular weekly resources, AI increasingly being used to expand ideation and project management capabilities (Figure 1). For example, as early as 2025, researchers documented that LLMs were capable of generating research ideas that were judged more novel (but slightly less feasible) than human experts[4]. The performance of these LLMs can be further improved via causal graphs. More specifically, LLMs can efficiently extract causal relation pairs from published articles. This allows the subsequent deployment of algorithms

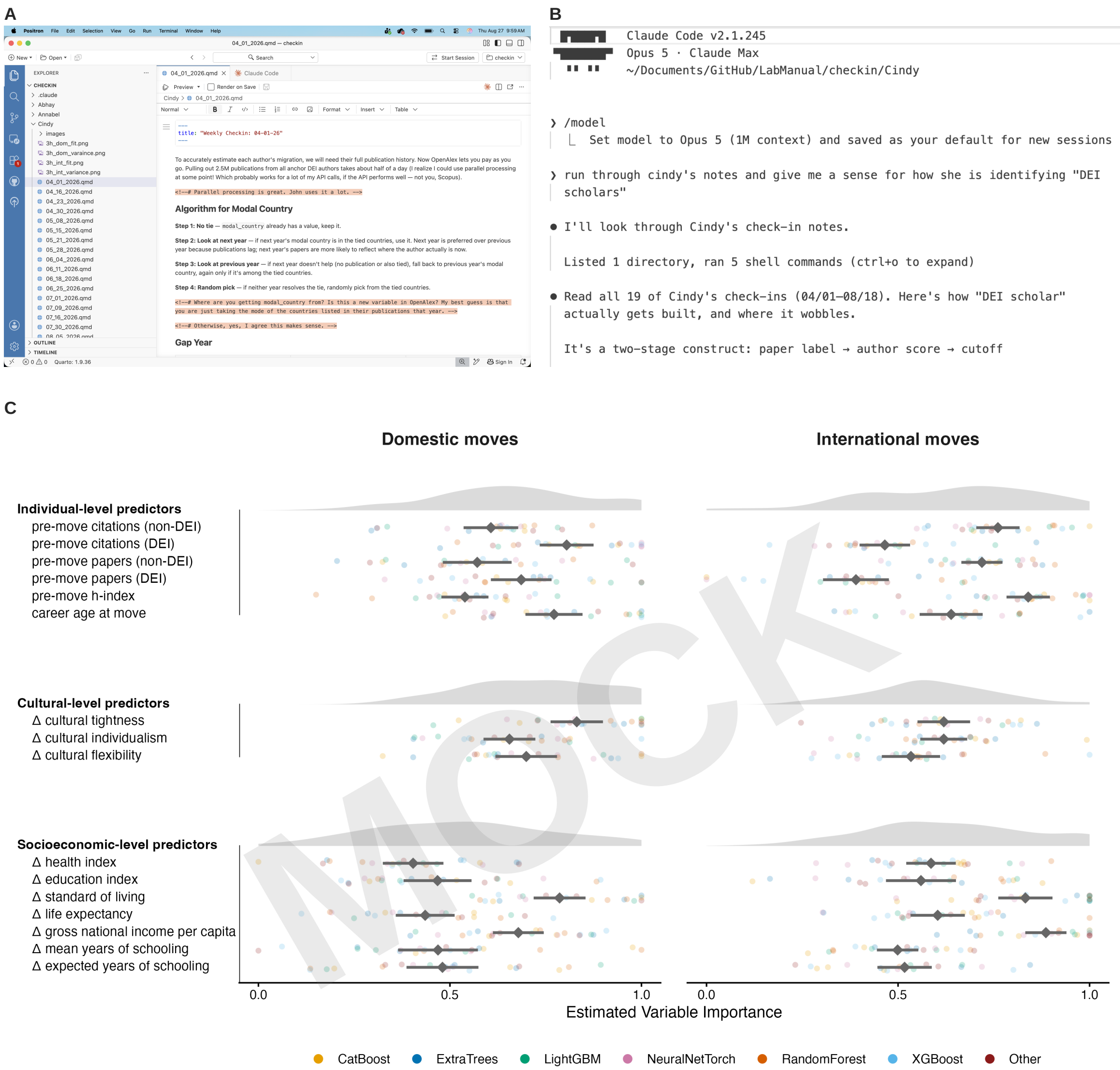

Figure 2: A seminar participant shared an LLM-assisted workflow to compile past meeting notes from their trainee (Panel A), review progress (Panel B), and mock a figure illustrating future directions (Panel C).

designed to predict missing links in this causal chain[5]. Similarly, LLMs can be used to *generate* theories based on pre-existing swaths of qualitative data[6] – an AI-for-data-analysis theme we continued to explore later in the seminar (see Chapter).

Areas of consensus – and dissensus

As expected, seminar discussion was marked by both areas of consensus and dissensus. In terms of consensus, there was broad agreement that LLMs can assist with ideation, project management, and the organization of qualitative sources of information. There was also broad agreement that this is changing the nature of competition in science – placing more emphasis on who can better leverage these tools for ideation and execution. Yet, there was clear disagreement about how far things should go. Seminar participants strongly disagreed about whether AI is reliably superior at ideation, whether the institution of science should shift to more AI-led work, and whether humanity should strive for fully autonomous AI scientists. These disagreements echo throughout the scientific landscape, with recent back-to-back Nature pieces on both the capabilities[7] and limitations[8] of science with fewer humans and more AI. These limitations became the focus of the following week of the Fall 2026 Collective [Artificial] Intelligence Seminar (see Chapter).

Ideation and Project Management: Critiquing AI

Covered in Week 3 of the seminar (September 3, 2026). Estimated date of arrival: September 13, 2026.

Literature Review and Synthesis

Literature Review and Synthesis: Using AI

Covered in Week 4 of the seminar (September 10, 2026). Estimated date of arrival: September 20, 2026.

Literature Review and Synthesis: Critiquing AI

Covered in Week 5 of the seminar (September 17, 2026). Estimated date of arrival: September 27, 2026.

Data Collection and Generation

Data Collection and Generation: Using AI

Covered in Week 6 of the seminar (September 24, 2026). Estimated date of arrival: October 4, 2026.

Data Collection and Generation: Critiquing AI

Covered in Week 7 of the seminar (October 1, 2026). Estimated date of arrival: October 11, 2026.

Analysis and Modeling

Analysis and Modeling: Using AI

Covered in Week 8 of the seminar (October 8, 2026). Estimated date of arrival: October 18, 2026.

Analysis and Modeling: Critiquing AI

Covered in Week 9 of the seminar (October 15, 2026). Estimated date of arrival: October 25, 2026.

Writing and Communication

Writing and Communication: Using AI

Covered in Week 10 of the seminar (October 22, 2026). Estimated date of arrival: November 1, 2026.

Writing and Communication: Critiquing AI

Covered in Week 11 of the seminar (October 29, 2026). Estimated date of arrival: November 8, 2026.

Review, Outreach, and Dissemination

Review, Outreach, and Dissemination: Using AI

Covered in Week 12 of the seminar (November 5, 2026). Estimated date of arrival: November 15, 2026.

Review, Outreach, and Dissemination: Critiquing AI

Covered in Week 13 of the seminar (November 12, 2026). Estimated date of arrival: November 22, 2026.

Personal AI Philosophy Statements

Covered in Week 14 of the seminar (November 19, 2026). Estimated date of arrival: November 29, 2026.

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